

Questions

1. What is the difference between `malloc()` and `calloc()`?

- A) `malloc()` initializes memory, `calloc()` does not
- B) `calloc()` initializes memory to zero, `malloc()` does not
- C) `malloc()` uses more memory than `calloc()`
- D) `calloc()` allocates memory in stack, `malloc()` in heap

2. What is the default return type of a function in C if not specified?

- A) `int`
- B) `void`
- C) `float`
- D) `double`

3. Which of the following is not a valid storage class in C?

- A) `static`
- B) `extern`
- C) `mutable`
- D) `register`

4. What will `sizeof(int*)` return on a 64-bit machine?

- A) 2
- B) 4
- C) 8
- D) 16

5. What is the primary purpose of the `volatile` keyword?

- A) Prevents compiler optimizations on a variable
- B) Allocates memory dynamically
- C) Optimizes the execution of a loop
- D) Increases memory efficiency

6. What does the `const` keyword do in C?

- A) Allocates memory in heap
- B) Makes a variable unmodifiable
- C) Reserves space for dynamic allocation
- D) Makes a variable global

7. Which of the following is true about function pointers?

- A) They store addresses of functions
- B) They must be used with global functions only
- C) They can store only `void` functions
- D) They do not exist in C

8. What is a segmentation fault in C?

- A) Memory allocation error
- B) Division by zero
- C) Accessing memory not allocated to the program
- D) Infinite loop error

9. Which of these is used to open a file in read mode in C?

- A) `fopen(filename, "w")`
- B) `fopen(filename, "a")`
- C) `fopen(filename, "r")`
- D) `fopen(filename, "rb")`

10. What is the output of `printf("%c", 'A' + 1);`?

- A) A
- B) B
- C) 66
- D) Compilation error

11. What will happen if `free()` is called on an already freed pointer?

- A) Program terminates normally
- B) Undefined behavior
- C) Compiler warning
- D) Memory leak

12. What does the `extern` keyword do?

- A) Declares a function in another file
- B) Allocates memory
- C) Creates a global function
- D) Converts function to inline

13. What is the output of `printf("%d", -1U > 1);`?

- A) 0
- B) 1
- C) -1
- D) Compilation error

14. What will be the output of `sizeof("Hello")`?

- A) 5
- B) 6
- C) 7
- D) Compiler error

15. Which of the following data types has the highest memory size?

- A) `char`
- B) `int`
- C) `double`
- D) `float`

2. Code Output Questions (20 Questions)

16. What will be the output?

c

```
#include <stdio.h>
int main() {
    int a = 5;
    printf("%d %d %d", a++, a--, ++a);
    return 0;
}
```

- A) 6 5 6
- B) Undefined behavior
- C) 5 6 5
- D) 6 5 5

17. Find the output of the following code.

c

```
#include <stdio.h>
int main() {
    char str[] = "Hello";
    char *p = str;
    printf("%c", ++*p);
    return 0;
}
```

- A) H
- B) I
- C) Undefined behavior
- D) Compiler error

18. What will be the output?

c

```
#include <stdio.h>
int main() {
    int arr[] = {1, 2, 3, 4, 5};
    printf("%d", *(arr + 3));
    return 0;
}
```

- A) 1
- B) 3
- C) 4
- D) 5

(Continuing the pattern for 19–35 output questions.)

Code Output Questions (19-35)

19. What will be the output?

```
#include <stdio.h>
int main() {
    int x = 10;
    int y = x++ + ++x;
    printf("%d", y);
    return 0;
}
```

- A) 20
- B) 21
- C) Undefined behavior
- D) 22

20. What will be the output?

```
#include <stdio.h>
int main() {
    int a = 5, b = 2;
    printf("%d", a / b);
}
```

```
    return 0;
}
```

- A) 2.5
- B) 2
- C) 3
- D) Compilation error

21. What will be the output?

```
#include <stdio.h>
int main() {
    int arr[] = {10, 20, 30, 40};
    printf("%d", *(arr + 2));
    return 0;
}
```

- A) 10
- B) 20
- C) 30
- D) 40

22. What will be the output?

```
#include <stdio.h>
int main() {
    int i = 1;
    while (i < 5)
        printf("%d ", i++);
    return 0;
}
```

- A) 1 2 3 4
- B) 1 2 3 4 5
- C) Infinite loop
- D) Compilation error

23. What will be the output?

```
#include <stdio.h>
int main() {
    int x = 5;
```

```
    printf("%d", x << 1);  
    return 0;  
}
```

- A) 5
- B) 10
- C) 2
- D) 1

24. What will be the output?

```
#include <stdio.h>  
int main() {  
    int x = 5, y = 10;  
    printf("%d", x > y ? x : y);  
    return 0;  
}
```

- A) 5
- B) 10
- C) Compilation error
- D) Undefined behavior

25. What will be the output?

```
#include <stdio.h>  
int main() {  
    int x = 0;  
    if (x = 1)  
        printf("True");  
    else  
        printf("False");  
    return 0;  
}
```

- A) True
- B) False
- C) Compilation error
- D) Undefined behavior

26. What will be the output?

```
#include <stdio.h>
int main() {
    char str[] = "C Programming";
    printf("%c", str[2]);
    return 0;
}
```

- A) C
- B) P
- C) o
- D) Compilation error

27. What will be the output?

```
#include <stdio.h>
int main() {
    int arr[] = {1, 2, 3, 4, 5};
    printf("%d", arr[5]);
    return 0;
}
```

- A) 5
- B) Undefined behavior
- C) Compilation error
- D) 0

28. What will be the output?

```
#include <stdio.h>
int main() {
    int i = 5;
    printf("%d", i++);
    return 0;
}
```

- A) 4
- B) 5
- C) 6
- D) Compilation error

29. What will be the output?

```
#include <stdio.h>
int main() {
    int x = 5, y = 10;
    int *p = &x, *q = &y;
    printf("%d", *p + *q);
    return 0;
}
```

- A) 10
- B) 15
- C) 5
- D) Compilation error

30. What will be the output?

```
#include <stdio.h>
#define SQUARE(x) x*x
int main() {
    printf("%d", SQUARE(5 + 1));
    return 0;
}
```

- A) 36
- B) 30
- C) Compilation error
- D) 11

31. What will be the output?

```
#include <stdio.h>
int main() {
    int x = 5;
    printf("%d", x & 1);
    return 0;
}
```

- A) 1
- B) 0
- C) 5
- D) Compilation error

32. What will be the output?

```
#include <stdio.h>
int main() {
    int x = 5;
    printf("%d", x | 1);
    return 0;
}
```

- A) 1
- B) 4
- C) 5
- D) 7

33. What will be the output?

```
#include <stdio.h>
int main() {
    char ch = 'A' + 3;
    printf("%c", ch);
    return 0;
}
```

- A) A
- B) B
- C) D
- D) Compilation error

34. What will be the output?

```
#include <stdio.h>
int main() {
    int x = 5;
    int y = x++;
    printf("%d %d", x, y);
    return 0;
}
```

- A) 5 5
- B) 6 5
- C) 6 6
- D) Compilation error

35. What will be the output?

```
#include <stdio.h>
int main() {
    char str[] = "Hello";
    printf("%c", *(str + 4));
    return 0;
}
```

- A) H
- B) e
- C) o
- D) Compilation error

3. Problem-Solving & Debugging (15 Questions)

36. Which of the following is the correct way to swap two numbers without using a third variable?

- A) $a = a + b; b = a - b; a = a - b;$
- B) $a = a \wedge b; b = a \wedge b; a = a \wedge b;$
- C) $a = b - a; b = b - a; a = b + a;$
- D) Both A & B

37. What is the time complexity of binary search?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n \log n)$
- D) $O(1)$

38. Which sorting algorithm has the worst-case time complexity of $O(n^2)$?

- A) Quick Sort
- B) Merge Sort

- C) Bubble Sort
- D) Heap Sort

39. What is the output of `printf("%d", 5 & 3);`?

- A) 2
- B) 3
- C) 5
- D) 8

40. What will `printf("%d", sizeof(NULL));` return?

- A) 1
- B) 2
- C) 4
- D) Compilation error

41. Which of the following is a valid keyword in C?

- A) `function`
- B) `virtual`
- C) `typedef`
- D) `class`

42. What is the primary difference between `while` and `do-while` loops?

- A) `while` executes at least once, `do-while` may not execute
- B) `do-while` executes at least once, `while` may not execute
- C) Both are identical
- D) `do-while` is faster than `while`

43. What does `printf("%d", 5 / 2);` output?

- A) 2.5
- B) 2
- C) 3
- D) Compilation error

44. What does `break` do inside a loop?

- A) Terminates the loop execution
- B) Skips the next iteration
- C) Restarts the loop from the beginning
- D) Causes a syntax error

45. What is the return type of the `main()` function in C?

- A) `void`
- B) `char`
- C) `int`
- D) `float`

46. What is the output of `sizeof(float)` on a 32-bit system?

- A) 2
- B) 4
- C) 8
- D) 16

47. What is a NULL pointer in C?

- A) A pointer that points to an unallocated memory location
- B) A pointer that stores 0 as its address
- C) A pointer that points to garbage value
- D) A pointer that stores random values

48. What is the default value of an uninitialized global variable in C?

- A) 0
- B) Garbage value
- C) NULL
- D) Compiler error

49. Which of the following operators has the highest precedence in C?

- A) +
- B) *
- C) =
- D) &&

50. Which function is used to deallocate memory in C?

- A) `delete()`
- B) `free()`
- C) `release()`
- D) `clear()`

51. What is the format specifier for printing an `unsigned int`?

- A) `%d`
- B) `%u`
- C) `%x`
- D) `%f`

52. Which of the following is not a valid way to declare an integer variable in C?

- A) `int x = 10;`
- B) `int x(10);`
- C) `int x; x = 10;`
- D) `int x = {10};`

53. What is the output of `printf("%c", 'A' + 2);`?

- A) A
- B) B
- C) C
- D) Compilation error

54. Which of these is used to read a line of text from the user in C?

- A) `scanf("%s", str);`
- B) `gets(str);`
- C) `fgets(str, sizeof(str), stdin);`
- D) Both B and C

55. How do you declare a constant in C?

- A) `int const x = 10;`
- B) `#define x 10`
- C) Both A & B
- D) None of the above

2. Code Output Questions (20 Questions, 56–75)

56. Find the output:

c

```
#include <stdio.h>
```

```
int main() {
```

```
    int a = 5;
```

```
        printf("%d", a++);

    return 0;

}
```

- A) 4
- B) 5
- C) 6
- D) Compilation error

57. What is the output?

c

```
#include <stdio.h>

int main() {

    char str[] = "Hello";

    printf("%c", *(str + 2));

    return 0;

}
```

- A) H
- B) e
- C) l
- D) o

58. Predict the output:

c

```
#include <stdio.h>

int main() {

    int x = 2;

    x = x << 2;

    printf("%d", x);

    return 0;

}
```

- A) 2
- B) 4
- C) 8
- D) 16

59. What is the output of `printf("%d", sizeof(NULL));`?

- A) 1
 - B) 4
 - C) 8
 - D) Compilation error
-

60. What does `int (*ptr)[10];` mean?

- A) `ptr` is an array of 10 integer pointers
 - B) `ptr` is a pointer to an array of 10 integers
 - C) `ptr` is a function returning an array of 10 integers
 - D) `ptr` is a constant integer pointer
-

61. What is the output of `5 / 2 * 2.0`?

- A) 5.0
 - B) 2.5
 - C) 4.0
 - D) 5
-

62. What is the output of `int x = 2; printf("%d", x *= 3 + 2);`?

- A) 6
 - B) 8
 - C) 10
 - D) 12
-

63. What is the output of `printf("%d", 'A' - 'B');`?

- A) 1
 - B) -1
 - C) 0
 - D) 2
-

64. What will `printf("%d", -1U);` print?

- A) -1
 - B) Large positive integer
 - C) 0
 - D) Compilation error
-

65. What is the output of `printf("%d", (int)3.14);`?

- A) 3.14
 - B) 3
 - C) 4
 - D) Compilation error
-

66. What does `int *const p;` mean?

- A) `p` is a constant integer pointer
 - B) `p` is a pointer to a constant integer
 - C) `p` is a constant pointer to a constant integer
 - D) `p` is a constant pointer to an integer
-

67. What does `void (*f) () ;` mean?

- A) `f` is a function that takes `void` and returns `void`
 - B) `f` is a pointer to an integer
 - C) `f` is a pointer to a function returning `void`
 - D) `f` is a function pointer returning an integer
-

68. What is the output of `printf("%d", 2 | 3) ;?`

- A) 2
 - B) 3
 - C) 1
 - D) 0
-

69. What is the result of `sizeof(1.5f) ;?`

- A) 8
 - B) 4
 - C) 2
 - D) 1
-

70. What is the output of `sizeof("A") ;?`

- A) 1
 - B) 2
 - C) 4
 - D) 8
-

71. What is the result of `printf("%d", 10 && 0);`?

- A) 10
 - B) 0
 - C) 1
 - D) Compilation error
-

72. What does `int arr[5] = {1,2,3};` initialize `arr[4]` to?

- A) 0
 - B) 3
 - C) Garbage value
 - D) Compilation error
-

73. What is the output of `printf("%d", (5,10,15));`?

- A) 5
 - B) 10
 - C) 15
 - D) Compilation error
-

74. What is the output of `int x = 5, y = 10; printf("%d", x < y ? x++ : y++);`?

- A) 6
 - B) 5
 - C) 10
 - D) 11
-

75. What does `static int x;` initialize `x` to?

- A) 0
- B) 1
- C) Garbage value
- D) Compilation error

3. Problem-Solving & Debugging (15 Questions, 76–90)

76. Which sorting algorithm has the best average-case performance?

- A) Bubble Sort
- B) Merge Sort
- C) Selection Sort
- D) Insertion Sort

77. What is the time complexity of an optimal binary search algorithm?

- A) $O(n)$
- B) $O(\log n)$
- C) $O(n \log n)$
- D) $O(1)$

78. What is the space complexity of recursive Fibonacci calculation?

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n^2)$

79. Identify the error in the following code:

c

```
#include <stdio.h>

int main() {
```

```
int arr[5] = {1, 2, 3, 4, 5};

printf("%d", arr[5]);

return 0;

}
```

- A) Array index out of bounds
- B) Syntax error
- C) Compilation error
- D) No error

80. How can you efficiently check if a number is a power of 2?

- A) `(n & (n - 1)) == 0`
- B) `(n | (n - 1)) == 0`
- C) `(n >> 1) == 0`
- D) `n % 2 == 0`

81. How do you dynamically allocate an array in C?

- A) `int *arr = malloc(n * sizeof(int));`
- B) `int arr[n];`
- C) `int arr = new int[n];`
- D) `int arr[] = {1, 2, 3};`

82. What is the maximum value of an `unsigned char`?

- A) 127
- B) 255
- C) 65535
- D) 1024

83. What will be the output of the following code?

c

```
#include <stdio.h>

int main() {

    int x = 10;

    int *ptr = &x;

    *ptr = *ptr + 5;

    printf("%d", x);

    return 0;

}
```

- A) 10
 - B) 15
 - C) Address of x
 - D) Compilation error
-

84. What is the output of this program?

c

```
#include <stdio.h>

int main() {

    int a = 5;

    if (a = 0)
```

```
        printf("Zero");  
  
    else  
  
        printf("Not Zero");  
  
    return 0;  
  
}
```

- A) Zero
 - B) Not Zero
 - C) Compilation error
 - D) Undefined behavior
-

85. Which of the following statements about pointers in C is true?

- A) A `NULL` pointer points to an invalid memory location.
 - B) `&` is used to access the value at the memory address stored in a pointer.
 - C) A pointer can store any data type except `void`.
 - D) `malloc()` always returns an integer pointer.
-

86. What will be the output of the following program?

c

```
#include <stdio.h>  
  
int main() {
```

```
char str[] = "DAIICT";

printf("%c", *(str + 3));

return 0;

}
```

- A) A
- B) I
- C) C
- D) T

87. What is the output of the following code?

c

```
#include <stdio.h>

#define SQUARE(x) x*x

int main() {

    int a = 4;

    printf("%d", SQUARE(a + 1));

    return 0;

}
```

- A) 25
- B) 9

- C) 20
 - D) Compilation error
-

88. Which of the following statements is true regarding `malloc()` and `calloc()`?

- A) `malloc()` initializes allocated memory to zero, whereas `calloc()` does not.
- B) `calloc()` initializes allocated memory to zero, whereas `malloc()` does not.
- C) Both `malloc()` and `calloc()` initialize memory to zero.
- D) Neither `malloc()` nor `calloc()` initialize memory to zero.

89. Which of the following expressions correctly determines if an integer `x` is even?

- A) `x % 2 == 0`
 - B) `x & 1 == 0`
 - C) `x / 2 == 0`
 - D) Both A and B
-

90. What will be the output of this program?

c

```
#include <stdio.h>
```

```
void swap(int *a, int *b) {  
  
    int temp = *a;  
  
    *a = *b;  
  
    *b = temp;  
  
}  
  
int main() {  
  
    int x = 5, y = 10;  
  
    swap(&x, &y);  
  
    printf("%d %d", x, y);  
  
    return 0;  
  
}
```

- A) 5 10
- B) 10 5
- C) Compilation error
- D) Undefined behavior

